

Devesh Maheshwari

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EDUCATION

Purdue University – West Lafayette
Bachelor of Science – Electrical Engineering & Computer Science

May 2027
GPA: 3.98 / 4.00

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Go, Java, SQL, JavaScript/TypeScript, Assembly x86-64
Embedded Systems: Embedded C, STM32, RP2350B/Raspberry Pi, Arduino, Verilog, VHDL, RTOS
Protocols & Interfaces: I²C, SPI, UART, USB, PWM, ADC, Digital Signal Processing
HW + Simulation: Oscilloscopes, multimeters, logic analyzer, soldering, LTspice, KiCad, MATLAB/Simulink, Synopsys TCAD
Tools & Infra: Linux, Git, GitHub, AWS, S3, SQL, MongoDB, GitLab, Docker, CI/CD, Kubernetes

RESEARCH

Undergraduate Research – LLMs in Embedded Systems (RTOS) | Jan 2026 – Present

- Led a 5-person team to benchmark LLM performance on embedded systems related PRs from Zephyr, NuttX & RIOT repos.
- Designed PR-sourcing pipeline that mined and filtered merged GitHub PRs into a curated, test-backed benchmark task set.
- Built Dockerized mini-swe-agent harness validating model-generated patches via tests in QEMU/Renode hardware sims.
- Evaluated five LLMs on pass-rate & step-efficiency against SWE-bench; all dropped on embedded tasks (best 70.6%).

PROJECTS

Gesture-Controlled Game – Raspberry Pi Project Sep 2025 – Dec 2025

- Built Frogger-style arcade game on an RP2350, played via real-time hand swipes read by APDS-9960 gesture sensor over I²C.
- Ran gesture detection and the game's state machine on separate RP2350 cores, linked through the inter-core hardware FIFO.
- Rendered graphics to SPI LCD with PWM sound effects, hardware pause/resume buttons, persistent SD-card leaderboard.

Optical Heart Rate Sensor – Analog and Digital Design Project Oct 2025 – Nov 2025

- Built infrared PPG sensor with IR204 LED & PT204 phototransistor, selecting resistors for stable mV-level waveform.
- Assembled cascaded high-pass and low-pass active filters with total gain ~250, isolating 40–200 BPM signals from noise.
- Created comparator with 3V hysteresis to generate -5 to 5 V pulses, driving indicator LED for real-time heartbeat detection.

Lunar Lander Game – FPGA and Verilog Project Jul 2025 – Aug 2025

- Engineered modular Verilog components (ALU, memory, control, display units) for FPGA lunar lander simulation at 100 Hz.
- Devised 16-bit BCD arithmetic with adders/subtractors, enabling single-cycle updates for altitude, velocity, fuel, thrust.
- Verified implementation on hardware with 100+ test cases, driving 7-segment displays and LEDs for real-time user feedback.

Audio Equalizer Circuit – Circuit Design Project Apr 2025 – May 2025

- Designed 3-band equalizer using op-amp RC/LC networks to implement low, mid, high-pass filters for frequency isolation.
- Combined amplification and potentiometer-based volume control, validating frequency response against LTSpice simulations.
- Reduced noise and distortion through proper biasing and filtering, allowing stable ~±10 dB per-band gain control.

Linux Shell Project – Advanced Custom Bash Shell Jan 2024 – Mar 2024

- Constructed custom linux-like shell in C/C++ with Lex/Yacc, supporting subshells, loops, wildcards, pipes, I/O redirection.
- Configured modular process, environment manager with 10+ built-in commands & extensible scripting language features.
- Achieved broad feature coverage across 50+ detailed test cases, ensuring correctness of parsing, execution, and redirection.

EXPERIENCE

Palo Alto Networks – Software Engineering Intern Santa Clara, CA | May 2026 – Aug 2026

- Building an internal usage tracker and dashboard that surfaces adoption metrics across platform microservices and features.
- Designing a plug-and-play telemetry workflow so new services onboard to the tracking pipeline with minimal manual setup.
- Automating collection, aggregation, and visualization of service-usage data to streamline internal developer workflows.

StudioX-AI – AI / ML Intern Remote, San Jose, CA | Jun 2025 – Aug 2025

- Automated DevOps and customer support with AI workflows, boosting internal efficiency and saving 5-8 hrs weekly.
- Integrated GitLab, Docker APIs with Claude via YAML for CI/CD and user management, sanity checks, and image builds.
- Packed integrations into DevOps chatbot enabling 10+ operations through conversational interface, reducing manual effort.
- Streamlined customer email intake with Claude, semantically analyzing ~50 emails/month to generate tickets.

Sopra Steria – Applied AI Intern Noida, India | May 2024 – Jul 2024

- Delivered RAG search over employee PDFs, reducing lookup time and generating grounded answers in under 5s.
- Synthesized ingestion pipeline by parsing PDFs, chunking ~800 tokens, embedding with OpenAI, indexing in FAISS db.
- Connected FAISS retrieval with GPT-4 to synthesize answers from top passages, improving response accuracy to 90%.